

The Epistemological Asymmetry of Artificial Intelligence across Science and Mathematics

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ABSTRACT

Artificial Intelligence (AI) architectures are increasingly capable of contributing to scientific discovery across a broad range of experimental disciplines, particularly in physics, chemistry, biology and engineering. In these domains, machine learning models can easily assist with hypothesis generation, detailed pattern discovery and complex design spaces exploration. In strict contrast to this, comparable advances in mathematics remain somewhat limited, despite significant advancement in automated theorem proving and formal verification. This perspective argues that the contrast reflects a deeper epistemological asymmetry between empirical scientific discovery and mathematical arguments. Typically, empirical sciences allow graded evaluation signals—such as prediction error, likelihood or experimental agreement—which enable heuristic search procedures to navigate smooth optimization landscapes. Mathematical reasoning, however, requires exact and strict logical validity: symbolic derivations are either correct or incorrect. Consequently, proof discovery corresponds to a combinatorial search over symbolic sequences in which valid proofs occupy a sparse subset of the hypothesis space. Drawing on ideas from proof complexity theory, we describe a verification–search asymmetry in formal reasoning that helps explain this difference. Identifying this structural yet conceptual distinction indicates that progress in AI-assisted mathematics will likely depend on hybrid architectures that combine generative exploration with rigorous symbolic verification. The introduced verification–search asymmetry offers a conceptual explanation for why AI architectures excel in empirical scientific discovery while remaining comparatively limited in mathematical reasoning.

1 Introduction

ARTIFICIAL INTELLIGENCE (AI)¹ architectures based on large language models and transformer neural network² (LLMs) and autonomous agent^{3,4} are rapidly transforming scientific research^{5–7}. During the past decade or so, machine learning methods have been deployed in a wide spectrum of disciplines within science^{8–10}, technology^{11,12}, engineering^{13,14} and mathematics (STEM), helping with tasks ranging from prediction of molecular structures^{15,16} and material discovery¹⁷ to code synthesis¹⁸ and experimental design. These systems increasingly operate not only as predictive tools but as exploratory engines capable of proposing hypotheses, generating candidate models, and navigating vast combinatorial search spaces.

The success of AI-assisted discovery has been particularly striking in empirical sciences and engineering, where many problems involve complex systems that are difficult to analyze directly. For instance, recently, the authors in¹⁹ used Google Gemini Deep Think Large Language Models that assisted in accelerating quantum-physics-based mathematical exploration, thereby solving an open problem in theoretical physics. Similar advances have emerged in cell-free protein synthesis^{16,20,21}, scientific engineering design^{21–24}, where generative models and reinforcement-learning agents have demonstrated the ability to optimize complex systems whose governing dynamics are only partially understood.

However, this remarkable progress reveals an intriguing and subtle asymmetry. Although AI architectures increasingly contribute to discovery in numerous scientific domains (some of them already cited), comparable breakthroughs in mathematics remain rare. Recent advances show that modern AI architectures can solve many structured competition problems, such as those appearing in the International Mathematical Olympiad^{25,26}; impressive, these successes occur in respective domains where solution spaces are comparatively constrained. In contrast, research-level mathematical reasoning involves navigating combinatorial expanding spaces of symbolic derivations in which valid proofs occupy an extremely small subset.

Despite significant advances in automated theorem proving²⁷ and formal verification, current AI architectures seldom produce the kind of deep conceptual insight that characterizes the major developments in contemporary mathematics. This disparity raises a natural and deep question: **why do artificial intelligence architectures appear well suited for discovery in empirical sciences but struggle with the forms of reasoning central to mathematical research and advancement?** This question deserves careful examination. Motivated by this very perspective, we investigate the structural origins of this asymmetry. We argue that the contrast arises not primarily from limitations of the present-day machine learning or robust neural network architecture, but rather from deeper differences between the epistemic structures of scientific discovery and

mathematical proof. In empirical sciences (such as theoretical physics and engineering), candidate theories or models can be evaluated using approximate criteria—numerical simulations, experimental measurements, smooth observable analysis, or statistical goodness-of-fit. This allows heuristic search procedures to generate useful hypotheses even when the underlying reasoning is partially complete or incomplete as a whole. The reverse, *mathematical reasoning requires exact logical validity*; so, it is imperative that the proposed proof must satisfy strict deductive constraints, and the accuracy cannot be continuously graded or approximated experimentally.

These differing epistemologies²⁸, as we learn, lead to different subtle computational landscapes. Indeed, many problems in scientific discovery can be *cast* as either an optimization or inference task in which numerically approximated solutions are acceptable and can be efficiently evaluated. The mathematical proving theorem, by contrast, corresponds to a discrete search over structured symbolic²⁹ objects—formal proofs—whose validity must ultimately be verified through precise logical rules. As we show below, this distinction can be formalized using concepts from proof-complexity theory, which highlight a fundamental asymmetry between proof verification and proof discovery. Underlying this perspective, we argue that this difference reflects a structural property of reasoning systems themselves, which we describe as a verification–search asymmetry.

Understanding this asymmetry is increasingly compelling, crucial, and important as AI architectures become embedded in scientific workflows. Rather than interpreting the relative difficulty of mathematical reasoning for AI as a temporary limitation of current architecture, we propose that it reflects a deeper structural feature of formal reasoning systems. Recognizing this distinction may help guide the development of future AI architectures better suited for mathematical discovery, while clarifying the domains in which heuristic generative systems are most likely to accelerate scientific progress. In science, discovery *can* precede proof, but in math, proof *is* discovery.

2 Artificial Intelligence as an engine of scientific discovery

In present times, contemporary advances in artificial intelligence have transformed the landscape of scientific research. Large language models, generative architectures³⁰, and autonomous agent systems are increasingly capable of helping in tasks that were once considered deep creative components of scientific practice. These architectures can propose candidate hypotheses, design computational experiments, generate code, and synthesize information from vast bodies of literature. Hence, AI is rapidly evolving from a pattern recognition tool to a platform for exploratory reasoning within scientific workflows. The impact of these developments has been particularly visible in empirical sciences and engineering. We discuss some of the notable interventions by AI in these scientific workflows as follows.

In molecular biology^{16,31,32}, the ML architecture has demonstrated unprecedented accuracy in predicting protein structures from amino-acid sequences, dramatically accelerating the pace of biology. In materials science and chemistry, generative models have been employed to search immense chemical spaces³³ for candidate molecules and materials with desirable properties. Similar techniques are now widely used in physics-informed ML³⁴, where NNs³⁵ approximate solutions of complex differential equations that govern physical systems. Across these complex domains, AI architectures help researchers and practitioners navigate high-dimensional hypothesis spaces that would be prohibitively expensive to explore manually. Engineering disciplines³⁶ have also witnessed a surge of AI-assisted innovation. Generative design frameworks allow algorithms to explore vast design spaces in aerospace engineering³⁷, architecture, and electronics, producing candidate configurations that satisfy multiple performance constraints simultaneously. Reinforcement-learning agents have demonstrated the ability to control complex dynamical systems, optimize manufacturing processes, and even design components for microelectronic circuits. In many of these applications, the role of AI is not merely predictive but exploratory: algorithms generate candidate solutions that can subsequently be evaluated through numerical analysis, simulation, experimentation, or physical prototyping.

A common feature unites these diverse successes; in each case, the underlying scientific task can be framed as either an optimization or inference problem over a giant hypothesis space. Candidate models, molecular structures, or engineering designs can be evaluated using quantitative metrics such as simulation accuracy, experimental measurements, or performance objectives. Importantly, the evaluation of these candidates typically tolerates an approximation. A model may be useful even if it is imperfect; a design may be valuable even if it is not optimal. This tolerance for approximate correctness allows heuristic search strategies—including those employed by modern machine learning systems—to produce valuable discoveries.

The growing integration of AI within scientific practice, therefore, reflects a deeper computational property of many empirical research problems: useful solutions can often be discovered through iterative exploration guided by approximate evaluation. Large language models and autonomous agents are particularly well suited to such environments because they excel at generating diverse candidate hypotheses and navigating complex search spaces using statistical patterns extracted from data. However, the success of AI across empirical sciences is in sharp contrast to its comparatively limited impact in mathematics. Although automated reasoning systems and formal proof assistants have achieved impressive progress in verifying existing proofs and assisting human mathematicians, the generation of fundamentally new mathematical insights remains rare. The reasons for this disparity are not merely technological, but structural, arising from differences in the epistemological foundations of scientific discovery and mathematical proof.

3 Epistemologies of empirical science and mathematics

The uneven impact of artificial intelligence across STEM fields reflects a difference deeper than the model architecture or the scale of the data set. It arises from a divergence in the epistemological structure of the underlying disciplines. Although science, technology, engineering, and mathematics are often grouped together under a common institutional umbrella, the criteria by which knowledge is generated, evaluated, and accepted differ substantially across these domains. This distinction becomes especially visible when one asks why heuristic generative systems have become increasingly useful in empirical discovery while remaining comparatively limited in deep mathematical reasoning.

In empirical science, knowledge is typically mediated by model architecture. A scientific model need not be exact to be valuable; rather, its legitimacy is established through explanatory power, predictive success, and agreement with observation. Theories in physics, chemistry, engineering, technology and other similar scientific domains are therefore often assessed through a graded notion of adequacy. One model may outperform another without being final or complete. Scientific discovery thus proceeds in an environment where partial success is meaningful and where candidate hypotheses can be ranked continuously according to performance. This feature is crucial to understanding the success of AI architectures in empirical scientific research. The heuristic search thrives when the space of candidate solutions admits smooth or at least informative evaluation signals. A generated molecule can be scored on the basis of predicted binding affinity; a candidate material can be ranked on the basis of estimated conductivity or stability; and an engineering design can be tested against performance objectives under simulation. In each case, imperfect outputs remain useful because they provide constructive directional information. The approximate correctness therefore has epistemic value: it helps guide a further search.

Mathematics is *naturally* organized differently. In mathematics, the central object is not an empirical model, but a formal statement whose validity must be established deductively. A theorem is not accepted because it works well in examples, nor because it appears statistically plausible, nor because it performs well under simulation. It is accepted only when it is embedded within a proof whose logical steps are valid under an agreed formal framework. This gives mathematics a much sharper notion of correctness than that found in most empirical sciences. Between a valid proof and an invalid one, there is no continuum of acceptability. There may be intuition, plausibility, and heuristic evidence surrounding a statement, but unfortunately none of these constitutes mathematical truth. The contrast is therefore not only between the “easy” and “hard” reasoning, but also between two distinct epistemologies. In empirical discovery, one often searches for models that are progressively more useful. In mathematics, one searches for proofs that are exactly correct. The former admits approximate guidance; the latter ultimately demands an exact deductive closure. This difference has profound computational implications. Continuous or graded evaluation makes it possible to navigate large hypothesis spaces through heuristic optimization. By contrast, an exact closure proof search requires locating highly structured symbolic objects whose validity is discrete and brittle: a single incorrect step can invalidate an otherwise sophisticated derivation.

Historically, many of the most important advances in physics and engineering have emerged before their full mathematical formalization was in place. Productive use has often preceded complete rigor. By contrast, mathematics does not permit such a separation between successful use and valid justification. A conjecture may be numerically compelling, topologically natural, and supported by many special cases, yet it remains unresolved until an exact-closure proof is found. What distinguishes mathematics, then, is not simply abstraction, but the necessity of deductive closure.

This distinction also clarifies why language-model-based systems encounter a particular difficulty in mathematics. Such systems are optimized to generate plausible continuations within symbolic sequences. In scientific settings, plausibility can be surprisingly productive because the candidate output can be filtered later through simulation, experiment, or downstream evaluation. However, in mathematics, syntactic plausibility is a weak proxy for deductive validity. A sequence of statements may resemble a proof stylistically while failing completely as a proof logically. The gap between local plausibility and global validity is therefore especially severe in formal reasoning. The central claim of this perspective is that this epistemological difference induces a structural asymmetry in the computational character of discovery itself. Scientific discovery often permits heuristic progress under approximate evaluation. Mathematical discovery requires an exact symbolic certification. It must be noted that this distinction does not imply that mathematics is forever inaccessible to AI, nor that empirical science is computationally simple. It suggests instead that current AI architectures are naturally aligned with domains in which useful intermediate signals exist and poorly aligned with domains in which correctness is sparse, discrete, and globally constrained. The theorem in the next section formalizes one aspect of this asymmetry through the lens of a Cook-Reckow proof complexity system.

The conceptual structure underlying this asymmetry is illustrated in [Figure 1](#). Empirical scientific discovery typically operates in environments where candidate hypotheses can be evaluated using graded performance signals, which induce optimization landscapes that heuristic algorithms can explore effectively. Mathematical discovery, by contrast, requires exact logical validity, so proof discovery becomes a combinatorial search problem in which valid derivations remain sparse.

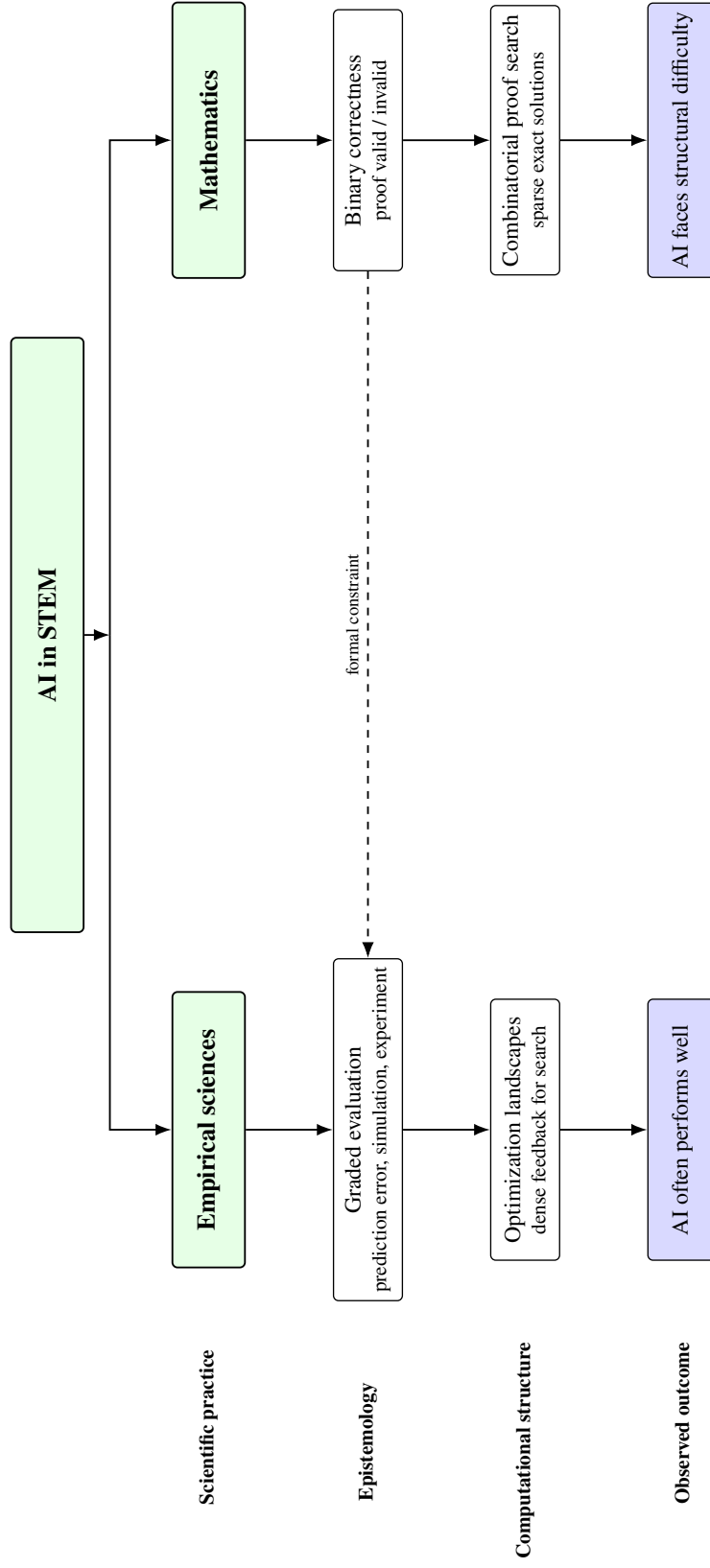


Figure 1. Structural layers underlying the asymmetry between AI-assisted discovery in empirical science and mathematics. Empirical sciences typically admit graded evaluation signals, allowing hypotheses to be ranked by predictive performance, simulation outcomes or experimental agreement. Such feedback induces optimization landscapes that heuristic systems can navigate effectively. Mathematics, by contrast, requires exact logical validity, so proof discovery becomes a combinatorial search over symbolic sequences in which valid proofs remain sparse. This structural distinction helps explain why AI systems often excel in empirical scientific tasks while encountering greater difficulty in deep mathematical reasoning.

4 Verification–search asymmetry in Mathematics

The preceding discussion suggests that the difficulty encountered by artificial intelligence architectures in mathematical discovery is not merely a consequence of current model limitations but may instead reflect a structural property of formal reasoning itself. This asymmetry can be clarified using ideas from proof complexity theory and propositional logic, particularly the framework introduced by Cook and Reckhow³⁸ for formal proof systems (CRP_fS).

A proof system can be viewed as a pair consisting of a formal language of statements and a verification procedure that determines whether a symbolic object constitutes a valid proof of a given statement. In the Cook–Reckhow formulation, a proof system is defined by a polynomial-time verification algorithm $V_{\text{poly}}(\phi, \pi)$ that accepts if and only if the string π is a valid proof of the formula ϕ . The key property is that verification is computationally efficient: once a candidate proof is provided, checking its correctness requires only polynomial time. This property, however, does not imply that discovering such proofs is computationally easy. The process of proof discovery *requires* searching through a space of symbolic derivations whose size typically grows combinatorial with respect to the depth of reasoning. If the branching factor of admissible symbolic transformations is denoted by b (non-zero) and the depth of reasoning by d , then the number of possible candidate derivations scales approximately as $\sim b^d$. Only a very small fraction of these symbolic sequences correspond to valid and acceptable proofs.

Theorem 1. (*Verification–search asymmetry*) *In any CRP_fS with polynomial-time verification, verification of a candidate proof can be performed efficiently, but the discovery of a valid proof generally requires a search over a symbolic space whose size grows combinatorial with the depth of reasoning. Consequently, the fraction of candidate symbolic sequences corresponding to valid proofs decreases rapidly as the proof length increases.*

Sketch. Let $V_{\text{poly}}(\phi, \pi)$ denote the polynomial-time verification algorithm for a proof system. Given a candidate proof π , the verifier checks the correctness of each inference step and confirms that the sequence of derivations terminates in the statement ϕ . Since this procedure operates in polynomial time complexity with respect to the length of π , the proof verification is computationally efficient. Proof discovery, on the contrary, can be modeled as a search process over sequences of symbolic transformations drawn from a finite rule set. If each step of the reasoning process admits approximately b admissible transformations, then the number of candidate derivations of length d increases on the order of b^d . Because only highly structured derivations correspond to valid proofs, the proportion of candidate sequences that satisfy the verification predicate $V_{\text{poly}}(\phi, \pi) = 1$ decreases rapidly as d increases. The result is a combinatorial search landscape in which valid proofs occupy a vanishingly small region of the overall symbolic space. $\square$

This asymmetry highlights a fundamental difference between mathematical reasoning and many empirical discovery tasks. In empirical sciences, approximate evaluation functions—such as prediction error or experimental agreement—provide continuous feedback that guides the search for promising regions of the hypothesis space. In formal mathematics, by contrast, the correctness is binary: a proof is either valid or simply invalid. As a consequence, heuristic exploration strategies that rely on graded feedback may struggle when applied to domains that require exact logical validity. Understanding this structural distinction is essential for interpreting the current capabilities and limitations of AI architectures in mathematical research. Rather than viewing the difficulty of theorem proving as a temporary deficiency of existing models, it may reflect an intrinsic property of symbolic reasoning systems in which verification is efficient but discovery remains challenging in the combinatorial sense. [Figure 2](#) and [Figure 3](#) illustrate this structural contrast. Although empirical discovery admits graded evaluation landscapes, formal proof discovery involves a combinatorial search over symbolic sequences in which valid proofs remain sparse.

This asymmetry also resonates with the classical intuitions of computational complexity theory in computer science. Numerous formal reasoning tasks exhibit the familiar structure in which verifying a candidate solution is computationally easier than discovering the solution itself. In formal proof systems, verification requires only checking that each inference step satisfies the rules of the formal calculus, whereas proof discovery requires navigating a combinatorial expanding space of symbolic derivations. Although the present argument does not rely on any particular complexity-theoretic separation, the verification–search asymmetry mirrors a broader computational pattern in which solution certification can be efficient even when solution discovery remains difficult.

5 Implications for Artificial Intelligence architectures

The verification–search asymmetry discussed above has important implications for the design of artificial intelligence architectures intended to assist in scientific reasoning. In many experimental domains, AI architectures benefit from the presence of continuous evaluation signals that provide informative gradients for exploration. The prediction error (standard loss functions), likelihood scores (log-likelihood), and experimental agreement offer graded quality measures that allow heuristic search procedures to identify progressively better hypotheses. Large language models and related generative systems perform (often exceptionally) well in such settings because approximate correctness still provides useful guidance for refinement.

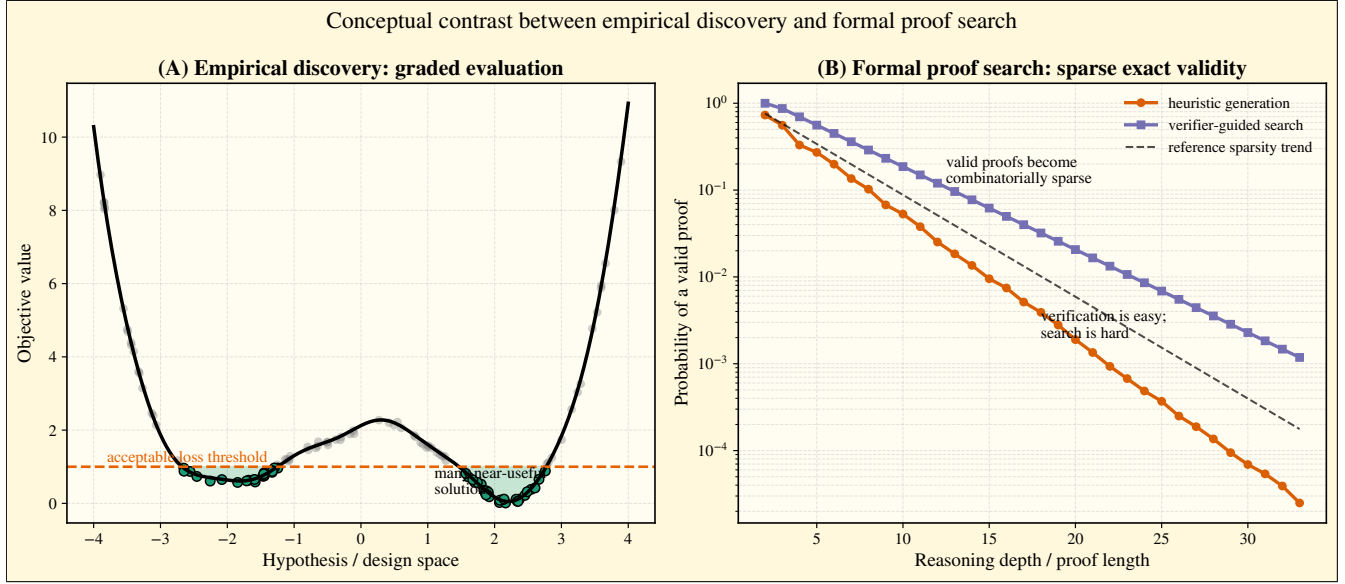


Figure 2. Conceptual contrast between empirical discovery and formal proof search. (A) In many empirical scientific domains, candidate hypotheses can be evaluated using graded performance measures such as prediction error or experimental agreement. As illustrated by the loss landscape, many hypotheses lie near acceptable performance thresholds, allowing heuristic search procedures to progressively improve candidate solutions. (B) In formal mathematics, by contrast, correctness is binary: a symbolic derivation is either a valid proof or just not. As reasoning depth increases, the space of candidate symbolic sequences grows rapidly while the fraction corresponding to valid proofs decreases sharply. Therefore, valid proofs become combinatorial sparse within the search space, illustrating the verification–search asymmetry discussed in the present section.

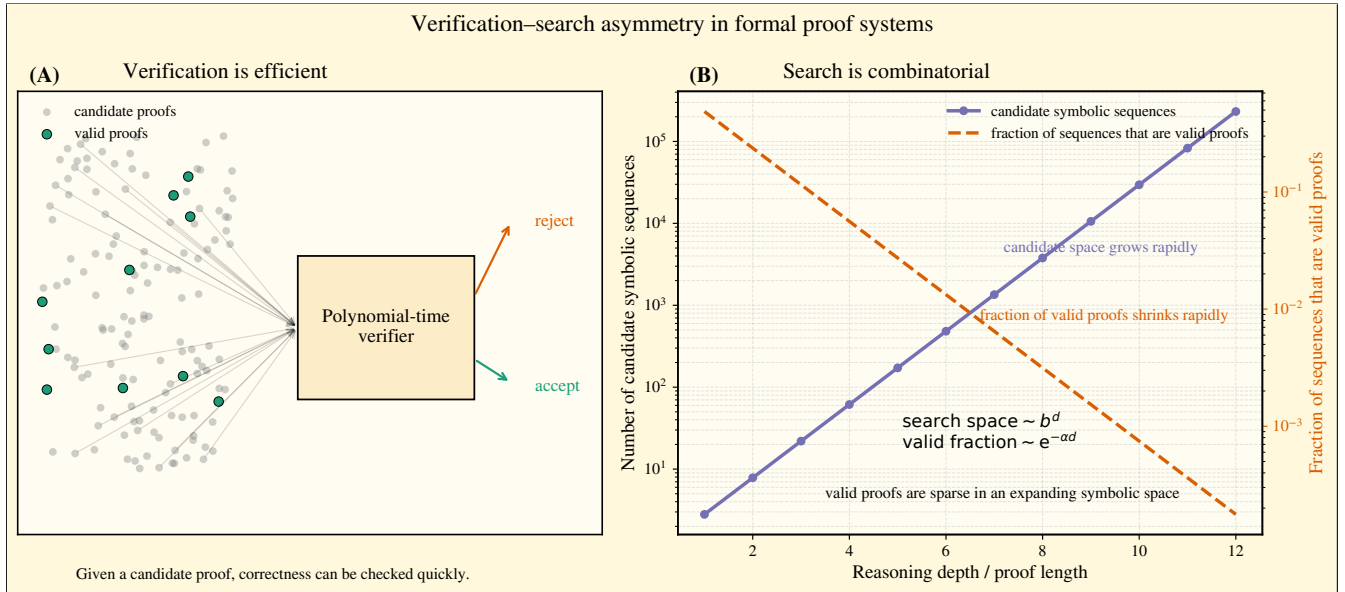


Figure 3. Verification–search asymmetry in formal proof systems. (A) In a CRP_fS , a candidate symbolic object can be checked efficiently by a polynomial-time verifier that determines whether the object constitutes a valid proof of a given statement. Verification therefore requires only local consistency checks of inference steps and can be performed in polynomial time with respect to the length of the candidate proof. (B) Proof discovery, however, on other hand, requires exploring a rapidly expanding space of symbolic derivations. As reasoning depth increases, the number of possible candidate sequences grows combinatorial while the fraction corresponding to valid proofs decreases sharply. Hence, valid proofs remain sparse within the symbolic search space, illustrating the structural asymmetry between efficient verification and difficult proof discovery discussed in this section.

Mathematical reasoning, on the contrary, lacks such intermediate qualitative feedback. A candidate proof is valid or invalid, and incorrect symbolic derivations provide little information about how they might be repaired. This absence of graded evaluation signals makes purely generative approaches to theorem discovery inherently difficult. Even if a language model produces syntactically plausible symbolic sequences, the probability that such sequences correspond to valid proofs remains extremely small within the combinatorial search space described in previous sections.

These observations help explain why current progress in AI-assisted mathematics often relies on hybrid architectures that combine generative models with formal verification systems. In recent theorem-proving frameworks, neural networks are typically used to guide the exploration of proof states, while symbolic proof assistants perform the final verification of logical correctness. Such systems effectively separate the tasks of exploration and validation: neural components propose candidate reasoning steps, whereas formal systems enforce the strict logical constraints of mathematical proof. This division of labor suggests a broader architectural principle for AI architectures engaged in formal reasoning. Rather than attempting to generate complete proofs in a single step, successful systems may instead operate through iterative interaction between heuristic proposal mechanisms and exact verification procedures. In this paradigm, generative models guide the search process by proposing promising symbolic transformations, while verification modules provide reliable feedback that restricts exploration to logically consistent regions of the search space.

More generally, overcoming verification-search asymmetry may require AI architectures that combine several complementary capabilities: symbolic reasoning engines capable of enforcing formal correctness, learning mechanisms that exploit structural regularities across numerous mathematical domains, and search strategies capable of navigating large combinatorial spaces. Such architectures resemble classical approaches to automated theorem proving but are augmented with modern machine learning techniques that improve heuristic guidance during proof exploration. These considerations suggest that progress in AI-assisted mathematics is unlikely to arise from generative language models alone. Instead, advances will likely emerge from hybrid-interaction systems that integrate statistical learning with formal symbolic reasoning. Understanding how to balance these components—combining the pattern-recognition abilities of neural networks with the exactness of logical inference—remains an open challenge at the intersection of artificial intelligence and mathematical logic.

6 Outlook

Artificial Intelligence is rapidly becoming an integral component in nearly every branch of scientific discovery. Across many empirical disciplines, including physics, chemistry, biology, and engineering, machine learning systems are increasingly capable of assisting in hypothesis generation, experimental design, and data interpretation. These successes have led to widespread expectations that similar advances will soon transform mathematical research. The analysis presented in this perspective indicates that the challenge of AI-assisted mathematics may arise not merely from limitations of current models or existing architecture but from deeper structural features of formal reasoning. Every mathematical proof has a vast degree of generality and requires exact symbolic validity, and the search for such proofs takes place within combinatorial expanding spaces of symbolic derivations. Since valid proofs occupy only a small subset of this space, the discovery process differs fundamentally from the optimization problems that characterize many empirical scientific tasks.

We emphasize that recognizing this distinction does not imply that AI architectures will remain limited in their ability to contribute to mathematics. Rather, it highlights the need for architectures specifically designed for formal reasoning. Hybrid systems that combine generative exploration with rigorous verification mechanisms may provide a promising path forward. Advances in formal proof assistants, neural-grounded search, and symbolic reasoning algorithms already suggest that such approaches can expand the scope of problems accessible to automated reasoning systems. At the same time, the asymmetry between verification and discovery can be construed as an opportunity rather than an obstacle. By clarifying the computational structure of mathematical reasoning, it becomes possible to design an AI architecture that takes advantage of this structure more effectively and efficiently. Understanding the verification-search asymmetry may therefore help guide the development of AI architectures specifically designed for formal reasoning, where generative exploration must be tightly coupled with rigorous symbolic verification.

Understanding the epistemological and computational distinctions between empirical discovery and formal reasoning can help guide the deployment of AI architectures in the sciences. Rather than expecting a single paradigm to perform uniformly across all domains of knowledge, it may be more productive to develop domain-aware architectures custom-tailored to the structure of the problems they address. Artificial Intelligence is unlikely to replace mathematical creativity. However, by illuminating the structure of the search spaces within which mathematical ideas emerge, AI architectures may increasingly serve as excellent collaborators that extend the reach of human reasoning. Exploring this possibility will require continued dialogue between mathematicians, computer scientists, and researchers in artificial intelligence—an interdisciplinary effort that may ultimately reshape the way mathematical discovery itself is conducted.

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Additional information

{Data and code} were not incorporated.